

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Hard

Question

$$\frac{7}{8}y - \frac{5}{8}x = \frac{4}{7} - \frac{7}{8}y$$
$$\frac{5}{4}x + \frac{7}{4} = py + \frac{15}{4}$$

In the given system of equations, p is a constant. If the system has no solution, what is the value of p ?

Correct Answer: 3.5, 7/2

Rationale

The correct answer is $\frac{7}{2}$. A system of two linear equations in two variables, x and y , has no solution if the lines represented by the equations in the xy -plane are distinct and parallel. Two lines represented by equations in standard form $Ax + By = C$, where A , B , and C are constants, are parallel if the coefficients for x and y in one equation are proportional to the corresponding coefficients in the other equation. The first equation in the given system, $\frac{7}{8}y - \frac{5}{8}x = \frac{4}{7} - \frac{7}{8}y$, can be written in standard form by adding $\frac{7}{8}y$ to both sides of the equation, which yields $\frac{14}{8}y - \frac{5}{8}x = \frac{4}{7}$, or $-\frac{5}{8}x + \frac{14}{8}y = \frac{4}{7}$. Multiplying each term in this equation by -8 yields $5x - 14y = -\frac{32}{7}$. The second equation in the given system, $\frac{5}{4}x + \frac{7}{4} = py + \frac{15}{4}$, can be written in standard form by subtracting $\frac{7}{4}$ and py from both sides of the equation, which yields $\frac{5}{4}x - py = \frac{8}{4}$. Multiplying each term in this equation by 4 yields $5x - 4py = 8$. The coefficient of x in the first equation, $5x - 14y = -\frac{32}{7}$, is equal to the coefficient of x in the second equation, $5x - 4py = 8$. For the lines to be parallel, and for the coefficients for x and y in one equation to be proportional to the corresponding coefficients in the other equation, the coefficient of y in the second equation must also be equal to the coefficient of y in the first equation. Therefore, $-14 = -4p$. Dividing both sides of this equation by -4 yields $\frac{-14}{-4} = p$, or $p = \frac{7}{2}$. Therefore, if the given system of equations has no solution, the value of p is $\frac{7}{2}$. Note that 7/2 and 3.5 are examples of ways to enter a correct answer.